DRUGS · HEAD AND NECK ONCOLOGY

# Pilocarpine and saliva substitutes

Dry mouth (xerostomia) is the most common permanent side effect of radiotherapy to the head and neck and is also the hallmark of Sjögren's syndrome. It causes difficulty eating, speaking and wearing dentures, altered taste, oral infections and rapid dental decay. Treatment uses salivary stimulants where gland tissue remains, saliva substitutes, and intensive dental prevention; prevention starts with parotid-sparing radiotherapy.

## 1 CHAPTER ONE At a glance

The drugs on one page.

### 1.1 Summary

PILOCARPINE AND SALIVA SUBSTITUTES

| | SUMMARY |
|---|---|
| **Pilocarpine** | Muscarinic agonist that stimulates remaining salivary tissue; 5 mg three times daily with or after meals, increased to a maximum of 30 mg a day; review at 12 weeks (benefit takes weeks) |
| **Cevimeline** | Muscarinic agonist used for Sjögren's syndrome (not available in the UK) |
| **Saliva substitutes** | Carboxymethylcellulose, mucin or xanthan gum sprays, gels and lozenges; neutral pH preferred to protect teeth |
| **Dental protection** | High-fluoride toothpaste (5,000 ppm), fluoride trays, regular dental review; dental assessment and extractions before radiotherapy |
| **Prevention** | Intensity-modulated radiotherapy sparing at least one parotid; amifostine (radioprotector) is effective but rarely used because of hypotension and nausea; submandibular gland transfer in selected patients |

# 2 CHAPTER TWO Using them in ENT

Practice.

## 2.1 After radiotherapy

Pilocarpine helps patients with residual gland function; it is less useful when glands are fibrotic. Sip water, sugar-free gum and humidification help. Watch for candidiasis and for **osteoradionecrosis** after dental extractions in irradiated bone.

# 3 CHAPTER THREE Safety

What goes wrong.

## 3.1 Pilocarpine adverse effects

Sweating (common), flushing, rhinitis, urinary frequency, diarrhoea, blurred vision (especially at night), bradycardia.

## 3.2 Contraindications

Uncontrolled asthma and COPD, narrow-angle glaucoma, acute iritis, significant cardiovascular disease, and beta blockers (conduction effects).

# 4 CHAPTER FOUR For the examination

What the examiner wants to hear.

## 4.1 Viva essentials

> WHAT THE EXAMINER IS LOOKING FOR

- Prevention of radiation xerostomia (parotid-sparing radiotherapy).
- Pilocarpine's mechanism, dose and contraindications.
- Dental care and osteoradionecrosis.

## 4.2 Common pitfalls

COMMON PITFALLS

- Stopping pilocarpine after 2 weeks for lack of effect.

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